

Multi-Theme Hardware Validation Plan

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Status: Validation package created. Hardware validation **NOT TESTED**. **Target:** `esphome/door_side_rotary.yaml` (Multi-Theme Firmware Engine) **Hardware:** ELECROW CrowPanel 1.28" ESP32-S3 Rotary Touch Display

This document provides the step-by-step plan for Hardik to physically validate the multi-theme firmware engine on the actual hardware.

See also: For the complete end-to-end project validation (including Raspberry Pi setup, Home Assistant, LocalTuya, bedside controller, and full system tests), refer to `end_to_end_project_validation_checklist.md`.

1. Flashing Guide

Before testing, the firmware must be flashed to the physical board.

1. Connect the ELECROW board to your computer via USB-C.
2. Ensure you have ESPHome installed locally or via Home Assistant.
3. Create or verify `esphome/secrets.yaml` exists with valid credentials.
4. Run the compile and upload command: `bash esphome run esphome/door_side_rotary.yaml`
5. Select the appropriate serial port when prompted.
6. Wait for the flash to complete and the device to reboot.
7. Keep the serial monitor open to observe boot logs and any immediate crashes.

2. Core Hardware Validation

TEST ITEM	EXPECTED BEHAVIOR	STATUS
Flash succeeds	ESPHome reports successful upload and device reboots.	PENDING
Device boots	Serial log shows successful boot without bootloops.	PENDING
Display renders	Screen turns on and displays the Power page.	PENDING
Touch works	Swiping left/right changes pages smoothly.	PENDING
Knob rotation works	Rotating the knob changes brightness (Page 2) or preset selection (Page 3).	PENDING

3. Interaction & Wake Behavior Validation

TEST ITEM	EXPECTED BEHAVIOR	STATUS
Screen sleep/idle	Screen turns off after 45 seconds of inactivity.	PENDING
Wake-only-first (Touch)	Touching a sleeping screen wakes it but does NOT trigger an action.	PENDING
Wake-only-first (Rotate)	Rotating knob while asleep wakes screen but does NOT change values.	PENDING
Wake-only-first (Press)	Pressing knob while asleep wakes screen but does NOT trigger action.	PENDING
Short press behavior	Pressing knob (<600ms) triggers page-specific action.	PENDING
Long press behavior	Holding knob (>1.5s) opens Theme Selector.	PENDING
Long press isolation	Long press does NOT also trigger the short-press action.	PENDING

4. Theme Selector Validation

TEST ITEM	EXPECTED BEHAVIOR	STATUS
Theme Selector opens	Long press transitions to Theme Selector page.	PENDING
Theme Selector browse	Rotating knob cycles through all 20 themes.	PENDING
All 20 themes selectable	Can reach Theme 01 through Theme 20.	PENDING

TEST ITEM	EXPECTED BEHAVIOR	STATUS
Theme Selector apply	Short press applies the selected theme and returns to Power page.	PENDING
Theme persistence	Active theme remains the same after a hard reboot.	PENDING

5. Page Functionality Validation

TEST ITEM	EXPECTED BEHAVIOR	STATUS
Power page works	Short press toggles <code>light.bedroom_group</code> power.	PENDING
Brightness page works	Rotation changes brightness; short press returns to Power page.	PENDING
Presets page works	Rotation highlights preset; short press applies it.	PENDING
4 locked presets work	Warm White, Soft Amber, Neutral White, Low Nightlight apply correctly.	PENDING

6. Home Assistant & Network Validation

TEST ITEM	EXPECTED BEHAVIOR	STATUS
HA target correct	All commands affect <code>light.bedroom_group</code> .	PENDING
HA diagnostic sensor	<code>text_sensor.veladial_doorside_active_theme</code> updates in HA.	PENDING
API unavailable state	Disconnecting WiFi/HA shows "Unavailable" badge on screen.	PENDING

7. Sensor & LED Validation

TEST ITEM	EXPECTED BEHAVIOR	STATUS
LED ring color	LED ring color matches the active theme's palette.	PENDING
TSL2591 adaptive backlight	(If connected) Display brightness adjusts to room ambient light.	PENDING
SHT45 diagnostic	(If connected) Temp/humidity report to HA but do not show on UI.	PENDING

8. Theme-by-Theme Visual Validation

Note: This checks if the theme engine successfully applies the distinct visual parameters for each theme. It does not demand visual perfection, only that the theme engine works.

THEME	EXPECTED VISUAL CHANGE	STATUS
01: Minimal Thermostat	Blue palette, minimal arcs	PENDING
02: SmartKnob-Inspired Arc	Purple palette, arc gauge layout	PENDING
03: Large Center Power Button	Green palette, large center button	PENDING
04: Single-Page Simple Mode	Warm amber palette, simplified layout	PENDING
05: Preset Ring UI	Classic amber palette, ring layout	PENDING
06: Night Mode Ultra-Minimal	Deep red palette, ultra-dim	PENDING
07: Text-First Utility	Amber palette, text-heavy	PENDING
08: Apple Watch Complications	Blue palette, complication layout	PENDING
09: LED-Ring Status-First	Amber palette, relies on physical LEDs	PENDING
10: Three-Screen Tab Carousel	Amber palette, tab carousel layout	PENDING
11: Brightness-First UI	Amber palette, brightness dominates	PENDING
12: Door Switch Replacement	Green palette, switch metaphor	PENDING
13: Lunar Phase Visualization	Moon white palette, lunar metaphor	PENDING
14: Sundial Shadow UI	Warm gold palette, shadow metaphor	PENDING
15: Tree Ring Growth Pattern	Wood tone palette, concentric rings	PENDING
16: Topographic Contour Map	Amber palette, contour lines	PENDING
17: Iris Aperture Mechanism	Amber palette, aperture mechanism	PENDING
18: Radar Sweep Animation	Amber palette, sweep animation	PENDING
19: Vinyl DJ Crossfader	Amber palette, turntable metaphor	PENDING
20: Eclipse Corona	Deep gold palette, corona glow	PENDING